Text Priors Transfer Little: Label Sharing and Test-Pool Calibration for Cold-Start Predictive AI Evaluation

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Abstract

The Predictive AI Evaluation Competition (PAIEC) scores predicted probabilities that an AI system answers a task correctly, on private benchmarks, first without labels and then with a few revealed labels per system and benchmark. We report four findings from building our entry. First, a prior learned from task and system text with a frozen multilingual encoder (E5) on 166 public benchmarks transfers almost nothing to held-out benchmarks, and its score, a weighted area under the Brier learning curve (ALC), stays within 0.13% of a Beta-binomial baseline that uses only the revealed labels. Second, sharing labels across systems on the same benchmark and across identical or similar items lowers ALC by 1.82% in a simulator of the official draws. Third, the official practice-phase (formative) test pool is more extreme than the training data. The feedback of a constant-prediction baseline reveals each group's distance from 50% accuracy, and reweighting simulated groups to match it moves the simulated score of this baseline from 0.2136 to 0.1826, against a mean of 0.1868 over 8 official runs. That feedback hides the side of 50%, but official runs of models that shift their predictions show it. Offsets from release date and model name helped in simulation, but on the platform the date offset raised the Brier score before any label by 0.026 and both by 0.035, which points to most test groups lying below 50% accuracy. Our final model drops these offsets and keeps a wide prior. On fresh simulated draws of recent systems it lowers ALC by 3.5% at the training side of 50% and by 2.9% to 3.1% at the side the official runs indicate, and without label sharing across systems it loses about half of that gain. Fourth, official runs of one baseline spread by 0.0493 over 8 draws and the leaderboard keeps each team's best run, so we compare submissions by repeated runs.

1 Introduction

In PAIEC a submission maps a description of an AI system (the *subject*) and a task (the *item*) to a probability of a correct answer [1]. The test benchmarks are private, so every prediction starts cold at the benchmark level. For each subject and benchmark pair (a *group*) the platform splits items into an acquisition pool and an evaluation pool and reveals acquisition labels at budgets of 0, 1, 3, 7, 15 and 31 labels. Each evaluation run scores a random *draw* of about nine groups (Section 2).

The obvious approach learns a prior from the text of items and subject descriptions on public benchmarks and adapts it to the revealed labels. Our earlier entries V2 and V3 followed it with frozen E5 embeddings and four or five training benchmarks [2]. Over 5 official runs, V2 averaged a Brier score of 0.2488 before any label was revealed, close to the 0.25 of a constant prediction of 0.5. We asked whether more data fixes this and trained on 166 binary benchmarks of measurement-db, about 36 times more recorded responses than V3. It did not. In leave-benchmark-out simulation the

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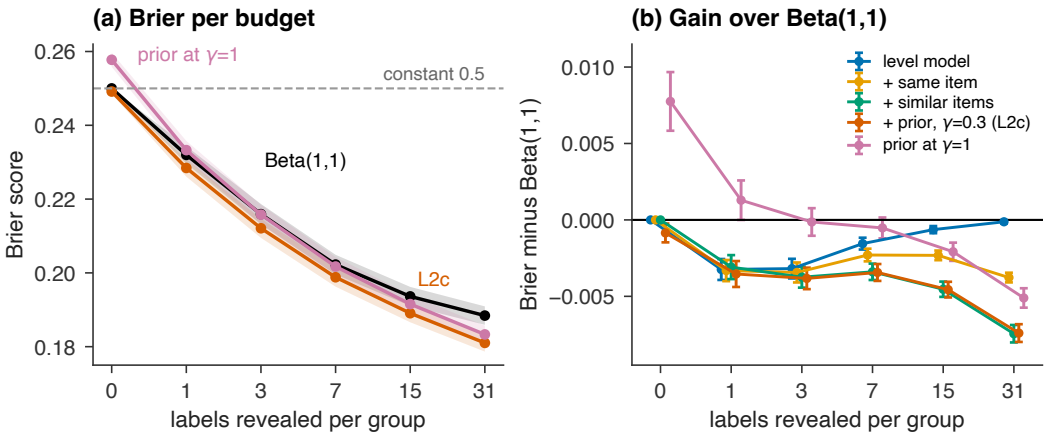


Figure 1: **On the training distribution, sharing labels across subjects and items carries the gain, and the text prior helps only at budget 0.** Simulator without test weights, all subjects, 600 fresh draws (4,823 groups), narrow level prior. (a) Mean Brier score per budget for Beta-binomial with a Beta(1,1) prior, the full narrow model (L2c) and L2c with the prior at full weight ($\gamma = 1$). (b) Paired difference from Beta(1,1) as each part is added. Bands and bars are 95% intervals over draws.

best prior weight γ was the smallest value we tried, and ALC matched the label-only Beta-binomial baseline within 0.13% (Section 7.1).

What helped was sharing label evidence (Section 4). The *level model* fits a benchmark offset from the labels of every subject in the draw. A same-item shift uses labels of other subjects on the same item, and a similar-item shift uses labels of items with close E5 embeddings. On fresh simulated draws this lowered ALC by 0.00388 (1.82%), and 95% of the gain remained without the text prior (Figure 1, Section 7.2).

Our simulator first missed the official level badly. It gave the official Beta-binomial baseline an ALC of 0.2136, while its 8 official runs averaged 0.1868. The reason is the test pool. Because the baseline predicts one constant per group, its official Brier and calibration scores reveal every group's distance from 50% accuracy, and test groups are far more extreme than training groups (Section 5). Reweighting simulated groups to this distribution reproduces the official baseline curve (Figure 2a). It also changes the best model. The narrow prior chosen before is no better than Beta-binomial under these weights, while a wider prior helps (Section 7.3). Offsets from release date and model name helped further in simulation, where recent training systems mostly score above 50%. On the platform the date offset raised the Brier score before any label by 0.026 (standard error 0.007). In the simulator this needs about 80% or more of test groups below 50%, and there L5 trails L3 by 0.0029 to 0.0047 ALC (Section 7.4). Our final model, L3, drops the offsets. It lowers ALC by 2.9% to 3.5% on fresh draws of recent subjects and changes little with the side of 50% (Figure 2b).

Official runs cannot confirm differences of this size one at a time. Our 8 runs of the Beta-binomial baseline spread by 0.0493, and the public leaderboard keeps each team's best run (Section 7.5). Our contributions are the following.

1. A measured negative result for frozen-encoder text priors, with two encoders and three training variants on 166 benchmarks (Section 7.1).
2. A label-sharing adaptation model that carries most of the attainable gain (Sections 4.2 and 7.2).
3. A calibration of the simulator to the formative test pool from aggregate feedback, the distance from 50% from a constant baseline and the side from models that shift their predictions. Each step changed our model choice (Sections 5, 7.3 and 7.4).
4. A repeated-run protocol for formative evaluation with a power calculation and an estimate of the best-of-N bias (Section 7.5).

Table 1: **V5 trains on about 36 times more recorded responses than V3.** Groups are subject-benchmark pairs. Accuracies are benchmark-weighted means over groups with at least 80 items.

	V3 training data	V5 corpus
Benchmarks	5	166
Recorded binary responses	1,142,294	40,775,314
Subject-item pairs and items		8,810,555 and 329,666
Subject configurations (with responses)		14,453 (11,420)
Groups with at least 80 items	82 (all sizes)	8,765 on 150 benchmarks
Mean group accuracy, all subjects		0.510 (55% above 0.5)
Mean group accuracy, released from July 2025		0.614 (68% above 0.5)

2 Task, score and the structure of official draws

A subject has eight string fields such as model name, harness and reasoning effort. An item has its content, metadata features, interactors and an anonymous benchmark alias. A submission's `predict([subject, item], labeled)` returns a probability, and `labeled` holds every label revealed so far in the draw, for all groups. For group g with Brier score $B_g(k)$ after k revealed labels,

$$\mathrm{ALC}_g = 0.1\, B_g(0) + 0.2\, \big[B_g(1) + B_g(3) + B_g(7) + B_g(15)\big] + 0.1\, B_g(31), \tag{1}$$

the normalized trapezoidal area under the Brier curve over $\log_2(1+k)$, and the score is the unweighted mean of ALC_g over the groups in the draw. $B_g(k)$ averages squared errors over all recorded responses of the group's evaluation items.

Our 44 formative feedback tables show the draw structure (Appendix E). Of these, 30 contain 9 groups on 7 benchmarks with 2 benchmarks contributing 2 subjects each, and 14 contain 8 groups on 7 benchmarks. The evaluation pool is half of a group's items and holds 44 to 76 items in all but 9 group evaluations, so most groups have about 88 to 152 items. All 382 group evaluations came from the same 7 benchmark aliases, and the curation guideline states that test subjects are systems from 2025 and 2026 [1].

Three assumptions link our simulations to this setting.

A1. Held-out public benchmarks with recent subjects, reweighted to the test pool's accuracy distribution (Section 5), stand in for the private benchmarks. If they differ in other ways, simulated gains need not transfer, in either direction.

A2. Two subjects drawn on one benchmark share most of their items, because the platform takes whole groups. The feedback reports only counts, so we cannot verify this. Without it, the same-item shift does nothing.

A3. A revealed label is one recorded response of the item. Our simulators draw it as a Bernoulli variable with the item's empirical success rate, which has the same distribution.

3 Data

All training data come from the measurement-db dataset at revision `54074453` [3], which is part of the competition's permitted inventory. Of its 281 benchmarks, 173 have binary responses. We did not download 4 of them (mind2web, naturalreasoning, swe_smith, tensortrust), which have one to three subjects each and include two with more than 500,000 items, and 3 more had no binary rows. The remaining 166 benchmarks form the corpus (Table 1). Each benchmark keeps at most 4,000 items, the ones answered by the most subjects, and we store the number of recorded trials n and successes k of every subject-item pair. A subject is the exact tuple of its eight fields, so one base model under another harness is a different subject. Subject descriptions are sparse. Of the 14,453 configurations, 45% have a model name, 31% a release date and 0.2% a reasoning effort.

4 Method

4.1 Text prior

Items and subjects are embedded with the frozen `intfloat/multilingual-e5-small` encoder [6] at revision `614241f`, with the recipe of V3 (Appendix B). An item text is its metadata line followed by its task, and a subject text lists its fields. A prior network gives a base logit z_{si} for subject s and item i. Two towers, each a two-layer perceptron with 256 Gaussian error linear units, map the subject input (E5 vector and release date) and the item input (E5 vector and text length) to h_s and h_i. The logit is $z_{si} = a(h_s) + b(h_i) + \langle Uh_s, Vh_i \rangle / \sqrt{32}$ with linear heads a, b and rank-32 projections U, V. Training minimizes the binomial log loss of the counts (n, k) of every pair, with weight $1/(nT_g)$ for a pair in group g with T_g trials. Each benchmark gets a free offset during training, and the mean offset is folded into the intercept afterwards.

4.2 Label-sharing adaptation

All labels in `labeled` that carry the target's benchmark alias enter the update, whichever subject they belong to. For one benchmark, the logit of subject s on item i is

$$\eta_{si} = \gamma z_{si} + o_s + c + d_s + u_i, \qquad c \sim \mathcal{N}(0, \tau_c^2), \quad d_s \sim \mathcal{N}(0, \tau_d^2), \tag{2}$$

where γ scales the text prior and o_s is a fixed subject offset (Section 4.3). The benchmark offset c is shared by all subjects in the draw, d_s is the offset of subject s, and u_i is an item shift. We fit c and all d_s jointly by Newton's method and take a Laplace approximation, which gives the posterior mean m_s and variance v_s of $c + d_s$. A label of another subject moves c and therefore the target's prediction. We call this part with $u_i = 0$ the *level model*.

The item shift uses residuals of the labeled pairs under the level model. For labeled pair j with subject s_j and item i_j, let $\hat{p}_j = \sigma(\gamma z_j + o_{s_j} + m_{s_j})$, $r_j = y_j - \hat{p}_j$ and $w_j = \hat{p}_j(1 - \hat{p}_j)$. For a kernel K with prior variance σ_K^2,

$$u_i = \sum_K \frac{\sigma_K^2 \sum_j K(i, i_j)\, r_j}{1 + \sigma_K^2 \sum_j K(i, i_j)\, w_j}, \tag{3}$$

which is one Newton step from zero for an offset shared by the target item and the labeled items, weighted by K. The same-item kernel $K(i, i') = \mathbf{1}[i = i']$ with variance σ_e^2 transfers labels of the same item from other subjects. The similar-item kernel $K(i, i') = \exp\big((\cos(e_i, e_{i'}) - 1)/h\big)$ on E5 vectors, with bandwidth h and variance σ_k^2, transfers labels of near-duplicate items. With $\eta^*_{si} = \gamma z_{si} + o_s + m_s + u_i$, the prediction $p_{si} = \sigma\big(\eta^*_{si} / \sqrt{1 + \pi v_s / 8}\big)$ integrates the posterior variance with the probit approximation [10].

4.3 Release-date and model-name offsets

The fixed offset o_s adds two subject priors fitted on training groups with at least 80 items. The date prior is $-0.472 + 0.285\, t$, with t the years from 2023-01-01 to the release date clipped to $[-3, 4]$, and -0.093 for an unknown date, fitted by benchmark-weighted least squares to group logit accuracy. The name ability of a normalized model name is the mean, over its training groups, of the group logit minus the benchmark's mean logit, shrunk with a pseudo-count of 2 and clipped to ± 2.5. It enters with weight 0.8, and the submission holds 683 names. In simulation the name ability comes from the four other benchmark folds only. There, it has a positive within-benchmark Spearman correlation with group accuracy on 71% of 62 held-out benchmarks with at least five named groups (median 0.18). L4 and L5 use these offsets, and the final model does not (Section 7.4).

4.4 Final submission

The final archive, L3, uses $\tau_c = 2.5$, $\tau_d = 1.0$, $\sigma_e^2 = 1$, $h = 0.03$, $\sigma_k^2 = 3$ and $\gamma = 0.3$, with $o_s = 0$. It holds the E5 weights, the prior network and the code (282.6 megabytes) and declares no `models.txt`. It runs on CPU with one prediction worker, which the platform restarts at every budget, so the encoder runs again for each budget. A draw of 8 groups then takes 8 to 20 minutes on a two-vCPU Colab CPU runtime (Appendix A). Caches of embeddings, logits and level fits are keyed by content hashes, so no prediction depends on earlier calls. No predictor reads any group identifier.

5 Calibrating the simulator to the formative test pool

For a group whose prediction is one constant q, the Brier score is $p(1-p)+(q-p)^2$ and the expected calibration error (ECE) is $|q-p|$, where p is the accuracy of the group's evaluation responses. The official Beta-binomial baseline predicts such a constant at every budget, namely 0.5 at budget 0. Its feedback therefore gives $|p-0.5| = \mathrm{ECE}_0$ for every group, and $p(1-p) = B_{31} - \mathrm{ECE}_{31}^2$ agrees with it to 10^{-6}.

Three baseline runs give 26 group evaluations. Their mean $p(1-p)$ is 0.136, against 0.184 for training groups, and the shares of $|p-0.5|$ in the bins $[0, 0.1)$, $[0.1, 0.2)$, $[0.2, 0.3)$, $[0.3, 0.4)$ and $[0.4, 0.5]$ are 0.115, 0.077, 0.231, 0.308, 0.269. So 58% of test groups have accuracy below 0.2 or above 0.8. The side of 0.5 is not identified, because flipping every label maps p to $1-p$ and q to $1-q$ and leaves both scores unchanged. Section 7.4 recovers it from models that shift their predictions.

Test weights reweight simulated groups so that their $|p-0.5|$ bins match these shares. The tuning grids used an earlier estimate (0.115, 0.077, 0.192, 0.308, 0.308) that placed one group in a different bin, and all confirmation runs use the shares above. With test weights the simulated official baseline, Beta(2,2), scores 0.1826 on all subjects and 0.1841 on recent subjects, against 0.2136 and 0.2105 without them. The three official runs average 0.1778 (standard error 0.0049), and the calibrated curve follows the official one at every budget (Figure 2a). Later runs, made after the model choice and not used for the weights, scored 0.2190, 0.1777, 0.1835, 0.2017, 0.1790, and all 8 runs average 0.1868 (standard error 0.0057). The matching curve is a consistency check, not a test of the model ranking, because once the shares of $|p-0.5|$ match it follows mostly from group sizes and the split. Official runs of V2 (mean 0.1825, 5 runs) and V3 (0.1862, 5) lie within one standard error of the Beta(2,2) mean, as the replay predicts (Section 7.3). Only such aggregate statistics and the per-archive scores of Section 7.4 enter model selection.

6 Evaluation protocols

Baseline. The Beta-binomial baseline predicts $(k+a)/(m+2a)$ for every evaluation item of a group with m revealed labels and k successes. Simulations use $a=1$ and the official archive uses $a=2$, which is worse by 0.00023 ALC in simulation (standard error 0.00019).

Leave-benchmark-out simulation. We split the 166 benchmarks into 5 random folds, train the prior on four and simulate on groups of the fifth with at least 80 items, subsampled to 80 to 160 items. Draws of the level model collect groups up to 1,000 items, 60 per fold, from 8,765 held-out groups.

Official-structure simulator. A draw picks 2 benchmarks whose two sampled subjects share 88 to 152 items and 5 benchmarks with one subject, and drops groups beyond 1,000 items, so most draws hold 8 groups. Each group is split in half independently. Base logits come from the prior trained without the pair's benchmark fold. Options restrict subjects to those released from 2025 (56 benchmarks) and apply test weights. Seeds 0 to 599 serve development, and 600 seeds from 100,000, 200,000 and 300,000 serve one confirmation run each. Standard errors treat the draw as the unit.

Runtime replays. We ran the real V2 and V3 submission code on the simulator's draws (recent subjects, seeds from 0) and paired each group with the simulator's L3, L5 and Beta-binomial rows. These draws exclude the nine benchmarks that V2 or V3 trained or tuned on.

Formative runs. Each Codabench submission is scored on a new random draw. We resubmit an uploaded archive several times and report the mean and standard error across runs.

7 Results

7.1 The text prior transfers almost nothing to held-out benchmarks

In all three training variants the best prior weight was $\gamma = 0.3$, the smallest value in the grid, and ALC fell steadily as γ shrank (Table 4 in Appendix C). Variant B, with free benchmark offsets during training, reached 0.21217 against 0.21244 for Beta-binomial, a difference of -0.13%. On a subset of

at most 600 items per benchmark, Qwen3-Embedding-0.6B [7] gave 0.21651 against 0.21684 for E5 and 0.21801 for Beta-binomial. One set of draws per run gives no standard error, so we read the gap between encoders as no evidence either way. The prior moves mainly budget 0. At $\gamma = 0.3$ it lowered the simulated budget-0 Brier score from 0.25 to 0.2492, and at full weight it raised it to 0.2579 (Figure 1a). V2's official runs averaged 0.2488. A zero-shot LLM judge did no better. Difficulty ratings by Qwen3-8B [8] improved simulated ALC by only 0.00035 and hurt budget 0, so the judge is not used (Appendix D).

7.2 On the training distribution, shared label evidence carries the gain

Without test weights, the narrow model ($\tau_c = 1.2$, $\tau_d = 0.6$, $\sigma_e^2 = 2$) lowered ALC from 0.21260 for Beta-binomial to 0.20872. This paired difference of -0.00388 (standard error 0.00023) comes from 600 fresh draws of all subjects. Without its text prior it reached -0.00369, which is 95% of the gain. Each level of sharing helps at a different budget (Figure 1b). The benchmark offset helps most at 1 to 3 labels, when the target subject has few labels but the other subject on the benchmark adds more. The same-item shift grows with the budget, and the similar-item shift doubles the gain at 31 labels.

7.3 On the test pool, prior width matters

Test weights change the ranking of models (Table 2). Without them the narrow model L2c is best in both subject populations, ahead of L3 by 0.0021 on all subjects. With them L3 beats L2c by 0.0023 on all subjects and 0.0014 on recent ones (standard errors 0.0002). Since only the weighted simulator reproduces the official baseline level, we read the unweighted ranking as a description of the training distribution, not of the test pool. The narrow level model with the same-item shift (L1c) is no better than Beta-binomial under test weights ($+0.0005$ on all subjects), although it gains 0.0029 without them. Extreme groups need large offsets, and a narrow prior holds their predictions too close to the starting value at low budgets.

On recent subjects with test weights, the wide model with item shifts and the E5 prior (L3) reaches -0.0063, or -3.5%, and its text prior contributes 0.0010 (standard error 0.0001). The date and name offsets lower ALC by a further 0.0010 (standard error 0.0001) to -0.0072 for L5, but only while the share of test groups below 50% stays under 0.47 (Section 7.4).

Two checks show that the gain on the test pool is not only a prior that suits extreme groups. First, the best Beta prior for this pool among those we tried, Beta(0.5,0.5), improves on Beta(1,1) by only 0.0006 under test weights on recent subjects, and L3 beats it by 0.0056 (standard error 0.0002). The official Beta(2,2) prior is worse than Beta(1,1) by 0.0055. Second, L3 restricted to the labels of the target group loses 0.0031 (standard error 0.0002) of its 0.0063 gain, because it no longer shares the benchmark offset or same items across subjects. In the wide level model, sharing only the benchmark offset is worth 0.0022 (standard error 0.0001) and needs no shared items (A2), and the same-item step adds 0.0006.

Replays of the real V2 and V3 code agree. Under test weights L3 beat them by 0.0045 and 0.0045 (standard errors 0.0013 and 0.0009), and without weights the differences were -0.0003 and $+0.0004$ (Appendix C).

7.4 Official runs reveal the side of 50%, which reverses the effect of the date and name offsets

Moving a group's constant prediction from 0.5 to $0.5 + s$ changes its Brier score by $s\,(s - 2(p - 0.5))$, so an upward shift raises the score unless $p > 0.5 + s/2$. The date and name offsets raise predictions for recent subjects, and test subjects are 2025 and 2026 releases. On the platform the offsets raised the budget-0 Brier score from 0.2489 for L3 (standard error 0.0013, 27 group evaluations) to 0.2746 for L4 (0.0065) and 0.2836 for L5 (0.0066), and every run of L4 and L5 lies above every run of L3 (Table 3). This is consistent with most test groups lying below 50% accuracy, while 62% of training groups of systems released from 2025 lie above it.

To size the effect we reweight simulated groups within each $|p - 0.5|$ bin so that a share π lies below 0.5. At the training share of 0.38 the simulated budget-0 gap between L4 and L3 is -0.003, and even at $\pi = 1$ it reaches only 0.023, below the official 0.026 (standard error 0.007). The one-sided lower 95% bound of the official gap needs $\pi \geq 0.80$ (two-sided, 0.74). No single share fits all three archives. The official L4 and L5 scores alone both point to π near 0.87, and the L4 score alone needs

Table 2: **Under test weights the wide models win, while without them the narrow model is best.** Official-structure simulator, the same 600 confirmation draws (seeds from 300,000) in each population. First row is the Beta(1,1) ALC, other rows are differences from it, lower is better. Groups keep the training side of 50%, and Figure 2b shows that the order of L3, L5 and L6 reverses when most lie below it. Rows with own labels only use the labels of the target group and share nothing across subjects. Standard errors are 0.0001 to 0.0003 per cell and near 0.0001 for paired differences between rows.

Subjects	all	all	recent	recent
Test weights	no	yes	no	yes
Beta-binomial ALC	0.2136	0.1769	0.2099	0.1786
Beta-binomial, Beta(0.5,0.5)	+0.0043	−0.0006	+0.0037	−0.0006
Beta-binomial, Beta(2,2), official prior	+0.0000	+0.0057	+0.0006	+0.0055
L1c, narrow level + same item	−0.0029	+0.0005	−0.0033	−0.0004
L2c, narrow, all sharing + E5 prior	−0.0043	−0.0027	−0.0062	−0.0049
Wide level model	−0.0002	−0.0027	−0.0008	−0.0032
+ same item	−0.0009	−0.0032	−0.0016	−0.0038
+ similar items	−0.0021	−0.0045	−0.0032	−0.0052
+ E5 prior, $\gamma = 0.3$ (L3, final)	−0.0023	−0.0049	−0.0039	−0.0063
+ date prior (L4)	−0.0023	−0.0052	−0.0043	−0.0068
+ name ability (L5)	−0.0021	−0.0051	−0.0044	−0.0072
L5 without the E5 prior	−0.0021	−0.0049	−0.0039	−0.0065
Wide level model, own labels only	+0.0017	−0.0009	+0.0013	−0.0010
L3, own labels only	+0.0009	−0.0023	−0.0003	−0.0031
L6, best of a grid on recent dev draws	−0.0023	−0.0045	−0.0051	−0.0075

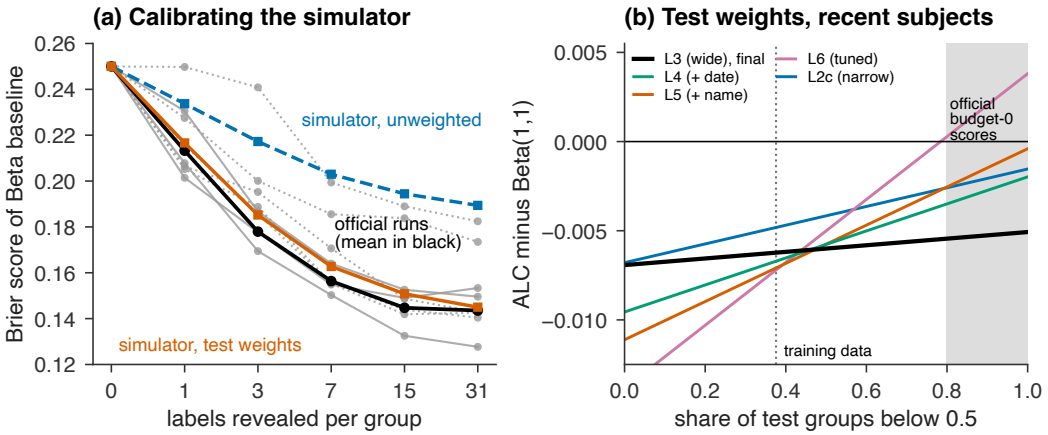


Figure 2: **Reweighting simulated groups to the test pool reproduces the official baseline curve, and the side of 50% decides whether date and name offsets help.** (a) Brier score per budget of the Beta(2,2) baseline in the three official runs that set the weights (grey, mean in black), in later runs (dotted) and in the simulator with and without test weights (all subjects, 600 confirmation draws). (b) Paired ALC difference from Beta(1,1) under test weights, recent subjects, 600 confirmation draws, as the share of groups below 0.5 varies within each $|p - 0.5|$ bin. Dotted line, training share. Shaded, shares allowed by the official budget-0 scores of L3 and L4. Standard errors are at most 0.0005.

$\pi \geq 0.70$, but the simulated L3 rises to 0.2554 at $\pi = 0.80$ against its official 0.2489, which alone points to 0.54. The gap between L4 and L3 removes the part they share, the text prior, so we use it. L5 beats L3 only for $\pi < 0.47$ (Figure 2b). L3 is better by 0.0029 at $\pi = 0.80$ and by 0.0047 at $\pi = 1$ (standard errors about 0.0001), and the tuned L6, with larger offsets, falls behind Beta-binomial for $\pi > 0.78$. The difference of L3 from Beta(1,1) stays between −0.0069 and −0.0051 over the whole range, and its official budget-0 score is below 0.25. On all subjects L5 and L3 differ by at most 0.0003, because older subjects get small offsets. Official mean ALC cannot decide it. L3 averages 0.1896, L4 0.1980 and L5 0.1927, but L3 is also behind Beta(2,2) at 0.1868 and V2 at 0.1825, all within the run-to-run standard deviation of 0.0161. We therefore submit L3.

Table 3: **Formative scores are single draws or small samples, but budget-0 scores separate the archives with date and name offsets.** Official Brier ALC per run, lower is better, each on its own draw of 8 or 9 groups, and the mean budget-0 Brier score $B(0)$. Weight runs set the test weights. L1c and L2c pool builds of one model for the CPU and GPU images.

Archive	Runs	ALC per run	Mean (s.e.)	$B(0)$
Beta(2,2), weight runs	3	0.187, 0.170, 0.177	0.1778 (0.0049)	0.250
Beta(2,2), later runs	5	0.219, 0.178, 0.183, 0.202, 0.179	0.1922 (0.0080)	0.250
V2	5	0.169, 0.179, 0.201, 0.164, 0.198	0.1825 (0.0075)	0.249
V3	5	0.171, 0.217, 0.186, 0.186, 0.171	0.1862 (0.0084)	0.243
L1c	8	0.180, 0.210, 0.192, 0.158, 0.205, 0.184, 0.199, 0.179	0.1884 (0.0060)	0.250
L2c	3	0.180, 0.213, 0.203	0.1986 (0.0099)	0.253
L3, final	3	0.205, 0.186, 0.177	0.1896 (0.0082)	0.249
L4	3	0.207, 0.207, 0.181	0.1980 (0.0087)	0.275
L5	7	0.177, 0.191, 0.216, 0.191, 0.180, 0.209, 0.184	0.1927 (0.0055)	0.284

7.5 Single formative runs cannot rank these methods

All 8 runs of the Beta(2,2) baseline archive have a standard deviation of 0.0161 (Table 3), and two other teams report 0.016 over 20 runs [4] and 0.024 over 4 runs [5]. Every run gets its own draw, so detecting a difference of 0.00388 with a two-sided 5% test and 80% power needs about 270 runs per archive, against 18 paired simulated draws. The leaderboard keeps each team's best run, which for 20 runs lies about 0.030 below the mean (one team reports a 20-run mean of 0.1765 and a leaderboard entry of 0.139 [4]), while the final evaluation scores each team once on a common set.

8 Related work

Item response theory models success through subject ability and item difficulty [11, 12], and tinyBenchmarks estimates a benchmark score from a few items with item parameters fitted on that benchmark [13], which a cold start lacks. Observational scaling laws predict scores from capabilities on known benchmarks [14], not item by item on unseen ones. Text-based difficulty estimation works within a domain [15], and amortized calibration predicts IRT difficulty from question content [18], the role of our text prior. Assessors predict a system's success on an instance [19], and a generic assessor does so for a new model tested on about 100 reference instances [20], while PAIEC gives at most 31 labels, all from the new benchmark. The level model is a random-intercept logistic model [16, 17].

9 Limitations

The main limit is the test-pool calibration. It rests on 26 group evaluations of three Beta runs and on the budget-0 scores of 3, 3 and 7 runs of L3, L4 and L5, all on the same 7 benchmark aliases, and model selection used it, so the formative pool is not untouched for our choice. The final evaluation draws a common subset from the same hidden data [1] but may weigh benchmarks differently. Resampling the bins of the 26 evaluations (20,000 replicates) keeps L3 ahead of Beta-binomial in every replicate and ahead of the narrow L2c in 85%, but ignores groups that recur across runs. Under the shares of the 44 less extreme evaluations of later Beta runs, the gaps of L3 to Beta(1,1) and to L2c are -0.0049 and $+0.0004$. If the final pool resembled training data, L2c would score about 0.0021 better than L3 (Table 2), and at the training side of 50% L5 would score about 0.0010 better. The same-item shift relies on assumption A2 and is worth up to 0.0006 under test weights. Confirmation draws correct for tuning but not for earlier design choices such as the simulator (Appendix A).

10 Conclusion

A frozen-encoder prior trained on 166 benchmarks changed ALC by 0.13% in leave-benchmark-out simulation, while sharing labels across subjects and items carried most of the gain, also against a Beta prior tuned to the test pool. Aggregate feedback calibrated our simulator to the test pool twice, with each group's distance from 50% accuracy from a constant baseline and its side from models that shift their predictions, and each step changed which model we submit.

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A Reproduction

Paths are relative to the project root. All CPU steps, the simulator and the runtime tests ran on the Mac with Python 3.12.12, torch 2.14.1, transformers 5.19.0, numpy 2.5.3, pandas 3.0.6, pyarrow 25.0.1 and huggingface_hub 1.33.0. GPU steps ran on Google Colab L4 runtimes with their preinstalled torch and transformers, driven by the colab command-line tool 0.7.4. Two mismatches remain between training and runtime. The prior's training weights give groups with repeated trials less weight than the score does, and training embeddings were stored in float16 while the runtime uses float32. The corpus omits 4 large or single-subject benchmarks and keeps the most-answered items of large benchmarks (Section 3). Because in-memory caches last one budget, L3 and L5 encode 405 to 588 texts per budget of a draw. On a two-vCPU Colab CPU runtime with one torch thread this costs 0.29 seconds per text and 4.5 seconds per model load, so a run reaches the 8-hour limit only at about 260 groups, against 8 or 9 in a formative run. **[TODO Add the public repository link.]**

1. Download and build the corpus (local CPU). The download fetches benchmark tables at the pinned revision and skips the four benchmarks listed in Section 3.

```
python - <<'EOF'
import json
from huggingface_hub import snapshot_download
R, REV = "aims-foundations/measurement-db", "54074453da7b9911be0f48e4c560be547bc84d21"
skip = {"mind2web", "naturalreasoning", "swe_smith", "tensortrust"}
keep = [b for b in json.load(open("colab_jobs/binary_benchmarks.json")) if b not in skip]
snapshot_download(R, repo_type="dataset", revision=REV, local_dir="data/mdb_full",
    allow_patterns=["*/formatted_tables/benchmarks.parquet", "*/formatted_tables/subjects.parquet"]
    + [f"{b}/formatted_tables/{t}.parquet" for b in keep for t in ("responses", "items")])
EOF
RAW=data/mdb_full OUT=data/corpus_full RUNTIME_DIR=v5 EMBED=0 SKIP_FETCH=1 python colab_jobs/build_corpus.
     py
```

2. Embeddings, priors and held-out logits (one L4 GPU). We run jobs on Colab through `colab_jobs/ctl.py` (budget check, upload, watch, download with SHA-256 check, stop) and `colab_jobs/supervisor.py` (status file, stall kill, release of idle machines). On the GPU machine, `/content/run` holds `v5/{spec,encoder,textproj,adapt}.py` and the job scripts, and `/content/work` holds the corpus.

```
python /content/run/embed_texts.py
python /content/run/train_eval.py cv --tag A
python /content/run/train_eval.py cv --tag B --bench-offset 1
python /content/run/train_eval.py cv --tag C --bench-offset 1 --dropout 0.3 --wd 0.01 --epochs 2
cd /content/run4 && RUNTIME_DIR=/content/run python export_z.py --tag B   # needs train_eval3.py in /
     content/run3
```

The last command writes `pairs_z.parquet` with held-out logits from five benchmark folds and `prior_B.npz`, the prior trained on all benchmarks (SHA-256 `486b104f...`). The encoder comparison runs `feat_bakeoff.py` and `train_eval2.py` with `-item-keys-file` on the 600-item subset. The subject offsets come from `python colab_jobs/fit_subject_priors.py`, which writes `date_prior.json`, `v5/name_ability.json` and `name_ability_folds.json` and reproduces our files exactly.

3. Test-pool statistics and simulations (local CPU). The first command computes the statistics of Section 5. The others reproduce Table 2, Figure 1, Figure 2b and Section 7.4, the runtime replays and the encoder cost. The baseline script reuses the ladder's draws, and `make_numbers.py` stops if they differ.

```
cd reports/paiec_v5_report
python -B scripts/test_pool_stats.py
python -B scripts/sim_ladder.py --tag recent_conf --min-date 2025-01-01 --seed0 300000  # also all_conf,
     and dev tags with --seed0 0
python -B scripts/sim_baselines.py --tag recent_conf --min-date 2025-01-01 --seed0 300000  # also all_conf
python -B scripts/sim_baselines.py --tag recent_conf --min-date 2025-01-01 --seed0 300000 --only beta,L3,
     L3_own \
  --out baselines_l3                                                                    # also all_conf
python -B scripts/side_of_half.py                                                     # Section 7.4
python -B scripts/sim_confirm.py --draws 600 --seed0 100000                           # Figure 1
python -B scripts/extra_stats.py                                                      # judge, name
     ability, replays
cd ../.. && EX=bertaqa,cellverse,cruxeval,engdesign,erbench,kernelbench,liveaopsbench,researchcodebench,
     tau2_bench
```

```
python -B colab_jobs/replay_runtime.py UNZIPPED_V2_DIR colab_jobs/results/replay_v2.parquet --seeds 0:120 \
  --min-date 2025-01-01 --exclude $EX
python -B reports/paiec_v5_report/scripts/record_encoder_texts.py UNZIPPED_L5_DIR texts.json
python reports/paiec_v5_report/scripts/time_encoder.py texts.json encode_timing.json  # CPU machine, next
     to v5/encoder.py
```

The last two commands give the encoder cost of Section 4.4. The first writes item texts from measurement-db, so its output is not in the repository. The ladder and confirmation scripts import frozen copies of the simulator (`scripts/sim_official_v2_snapshot.py`, SHA-256 `8da09880...`, and `scripts/sim_official_snapshot.py`). The paired simulator rows came from `sim_official.py -emb data/corpus_full/item_emb.npy -draws 120 -min-date 2025-01-01 -exclude $EX -dump colab_jobs/results/sim_dump_recent_ex.parquet` with the variants Beta, L1c, L3, L4 and L5. The dump stores raw per-group ALC and p, so test weights are applied afterwards.

4. Final archive. The first command rebuilds L3 and the last checks it with the validator of the organizers' starting kit, which is not part of our repository. The second builds L5.

```
P=colab_jobs/results/exportB/work/results
python v5/build_zip.py $P/prior_B.npz $P/prior_B.json 2.5 1.0 0.3 dist/v5/L3_wide_prior.zip \
  '{"use_prior": true, "item_exact": 1.0, "item_kernel": [0.03, 3.0]}'
python v5/build_zip.py $P/prior_B.npz $P/prior_B.json 2.5 1.0 0.3 dist/v5/L5_wide_date_name.zip \
  '{"use_prior": true, "item_exact": 1.0, "item_kernel": [0.03, 3.0], "name_scale": 0.8, "date_prior": '"$
     (cat colab_jobs/results/date_prior.json)"'}'
python starting_kit/tools/check_submission_zip.py dist/v5/L3_wide_prior.zip
```

Our L3 archive (Codabench submission 1033014) has SHA-256

`a47150effc0761d29b206ccb548d3c7da6a063e7a22072a352e60dc5b15f7ee7`.

We built it before the offset code entered `model.py`. A rebuild adds that code and the unused name table, and on 360 synthetic calls with 0, 5 and 30 labels it returns identical predictions. The L5 archive has SHA-256 `2d8b7034...` The script `build_zip.py` takes the five E5 files from the V3 archive or, if it is absent, from the hub at the pinned revision, and checks their SHA-256. A rebuild gives identical file contents.

5. Formative runs and this report. The first three commands resubmit an uploaded archive, collect the scores of finished runs and record the runs still waiting in the queue. The others rebuild the numbers, the figures and this PDF.

```
python scripts/resubmit.py L5_wide_date_name.zip 3 --label L5
python scripts/track.py
python scripts/queue_snapshot.py
cd reports/paiec_v5_report && python -B scripts/make_numbers.py && python -B scripts/make_figures.py
pdflatex main.tex && pdflatex main.tex
```

B Hyperparameters and prompts

Text projection. Each text gets the E5 query prefix, and hidden states are mean-pooled under the attention mask and normalized to unit length. Texts longer than 8,192 characters keep a head, a middle and a tail span. After tokenization, inputs longer than 510 tokens keep four spans of 128, 127, 127 and 128 tokens spread evenly over the text. A JSON chat transcript that ends in a user turn keeps up to 512 characters of system messages and the final user message. The item text is the first 300 characters of the item features followed by this projection. The subject text lists name, provider, release date, harness with version, reasoning effort and the first 300 characters of the extras.

Prior. Subject input is the E5 vector, the years from 2023 to the release date clipped to ± 5 and a missing-date flag. Item input is the E5 vector and $\log(1 + \text{characters})/10$. Hidden width 256, rank 32, dropout 0.1, AdamW [9] with learning rate 2×10^{-3} and weight decay 10^{-4}, one-cycle schedule with 10% warm-up, batch 8,192, 4 epochs, seed 99 for the final prior and the fold index for fold priors.

Table 4: **In every variant the best prior weight is the smallest one tried.** Leave-benchmark-out ALC with the level model, best (τ_c, τ_d) per row, lower is better. Each block has its own draws and baseline. Variant C adds dropout 0.3, weight decay 0.01 and 2 epochs to B.

Data	Prior	ALC	vs Beta-binomial
Full corpus	Beta-binomial, Beta(1,1)	0.21244	
	B, $\gamma = 0.3$	0.21217	-0.13%
	B, γ from 0.45 to 0.9	0.21258 to 0.21503	
	A (no benchmark offsets), $\gamma = 0.3$	0.21260	$+0.08\%$
	C (more regularization), $\gamma = 0.3$	0.21229	-0.07%
600-item subset	Beta-binomial, Beta(1,1)	0.21801	
	E5 prior, $\gamma = 0.3$	0.21684	-0.54%
	Qwen3-Embedding-0.6B prior, $\gamma = 0.3$	0.21651	-0.68%

Leave-benchmark-out grid. The grid crosses $\tau_c \in \{0.8, 1.2, 1.8, 2.5, 3.5\}$ with $\tau_d \in \{0.5, 0.8, 1.2\}$ and $\gamma \in \{0.3, 0.45, 0.6, 0.75, 0.9\}$, 75 configurations in total. The first run used γ from 0.7 upward and was 1.32% worse than Beta-binomial.

Simulator grids. Newton iterations stop after 25 steps or when the largest step is below 10^{-7}. All three grids of the official-structure simulator used 400 development draws and include the Beta(1,1) baseline. Here $\gamma = 0$ means no text prior. Grid 1 (no test weights, all subjects, 487 configurations) crossed $\tau_c \in \{0.9, 1.2, 1.6\}$, $\tau_d \in \{0.6, 0.8, 1.1\}$, $\sigma_e^2 \in \{2, 4, 8\}$, $h \in \{0.02, 0.03, 0.05\}$, $\sigma_k^2 \in \{1, 1.5, 3\}$ and $\gamma \in \{0, 0.3\}$. Its best difference was -0.00463 (standard error 0.00028), and it chose the narrow configuration. Grid 2 (earlier test-weight shares, all subjects, 289 configurations) crossed $\tau_c \in \{2, 2.5, 3.5, 5\}$, $\tau_d \in \{0.8, 1, 1.4, 2\}$, $\sigma_e^2 \in \{1, 2, 4\}$, $\sigma_k^2 \in \{1.5, 3, 6\}$ at $h = 0.03$, and $\gamma \in \{0, 0.3\}$. It chose the wide level prior of L3. Grid 3 (earlier test-weight shares, subjects from 2025, 326 configurations with L5) crossed $\tau_c \in \{2, 2.5, 3.2\}$, $\tau_d \in \{0.8, 1, 1.3\}$, $\sigma_e^2 \in \{0.5, 1, 2\}$, $\gamma \in \{0.2, 0.3, 0.45\}$, name weight $\{0.8, 1.2\}$ and date weight $\{1, 1.5\}$ at $\sigma_k^2 = 3$ and $h = 0.03$. Its best configuration is L6. Top lines are in `colab_jobs/results/sim_testw_grid2.log` and `sim_grid3_recent.log`. Of the 600 draws of the unweighted confirmation run, 119 hold 9 groups, 385 hold 8 and 96 hold 7.

Judge prompt. The system message reads "You estimate how hard a task is for current frontier AI systems. Read the task and rate its difficulty on a scale from 1 to 9, where 1 means almost every AI system solves it correctly and 9 means almost none does. Answer with a single digit only." The user message is "Task:" followed by the item text, truncated to 1,024 tokens of head and tail, and "Difficulty (1-9):". Thinking mode is off.

C Leave-benchmark-out results and runtime replays

With test weights, L3 beat the real V2 code by 0.0045 (standard error 0.0013, 120 draws, 974 groups) and V3 by 0.0045 (0.0009, 120 draws). Without weights the differences were -0.0003 (0.0009) and $+0.0004$ (0.0008). V2 was close to Beta-binomial under test weights (-0.0013, standard error 0.0013) and better without them (-0.0029). So V2 and V3 match L3 on the training distribution but fall behind it on the extreme test pool, where their narrow priors lose the gain over Beta-binomial. A replay of the L5 runtime on 3 draws (23 groups) matched the simulator within 0.0044 ALC per group on average, and a rebuild of L3 with that code returns identical predictions (Appendix A).

D The LLM difficulty judge

Following the use of LLMs as graders [21], Qwen3-8B [8] rated item difficulty from 1 to 9 as the expected digit of its next-token distribution, with the prompt of Appendix B. It rated 13,880 of 18,952 sampled items before we stopped the run to save compute, so long items are missing. The within-benchmark Spearman correlation between rating and item accuracy had median -0.062 and was negative, the useful direction, on 63% of 131 benchmarks. Between benchmarks it was -0.049. In the simulator on rated items with test weights, the best weight of 0.2 on the standardized rating improved ALC by only 0.00035 and hurt budget 0.

E Formative runs

Table 5 lists every finished formative run of our team. V1 and the V3 variant with an uncertainty band are earlier entries not described here.

Table 5: **All finished formative runs.** Means over groups of one draw each. B_0 and B_{31} are mean Brier scores at budgets 0 and 31.

Submission	Archive	Groups	Benchmarks	ALC	B_0	B_{31}
1003760	Beta(2,2) baseline	9	7	0.1867	0.2500	0.1496
1003791	V1 hierarchical prior	8	7	0.1911	0.2511	0.1612
1004759	V2	9	7	0.1695	0.2536	0.1330
1004796	V3	9	7	0.1711	0.2401	0.1478
1004970	V3 with uncertainty band	9	7	0.1876	0.2341	0.1644
1032873	V2	8	7	0.1790	0.2480	0.1484
1032874	V2	9	7	0.2013	0.2472	0.1699
1032875	V2	8	7	0.1644	0.2452	0.1303
1032876	V2	9	7	0.1983	0.2502	0.1802
1032877	V3	9	7	0.2172	0.2709	0.1843
1032878	V3	9	7	0.1861	0.2421	0.1594
1032879	V3	9	7	0.1857	0.2300	0.1565
1032880	V3	9	7	0.1709	0.2297	0.1396
1032881	Beta(2,2) baseline	9	7	0.1698	0.2500	0.1277
1032882	Beta(2,2) baseline	8	7	0.1769	0.2500	0.1533
1032902	L1c, GPU image	9	7	0.1803	0.2500	0.1528
1032903	L1c, GPU image	8	7	0.2102	0.2500	0.1815
1032904	L1c, GPU image	8	7	0.1919	0.2500	0.1627
1032928	L2c, GPU image	9	7	0.1795	0.2529	0.1496
1032929	L2c, GPU image	8	7	0.2128	0.2548	0.2006
1032930	L2c, GPU image	8	7	0.2034	0.2527	0.1736
1032960	L1c	9	7	0.1578	0.2500	0.1123
1032962	L1c	8	7	0.2052	0.2500	0.1819
1032963	L1c	9	7	0.1840	0.2500	0.1465
1033014	L3, final	9	7	0.2053	0.2511	0.1856
1033015	L3, final	9	7	0.1860	0.2491	0.1661
1033016	L3, final	9	7	0.1775	0.2466	0.1451
1033024	L4	8	7	0.2066	0.2723	0.1703
1033025	L4	9	7	0.2068	0.2768	0.1735
1033026	L4	9	7	0.1807	0.2744	0.1528
1033031	L5	9	7	0.1774	0.2870	0.1473
1033033	L5	9	7	0.1915	0.2878	0.1550
1033034	L5	9	7	0.2165	0.2790	0.1771
1034238	L5	8	7	0.1909	0.2911	0.1261
1034630	Beta(2,2) baseline	9	7	0.2190	0.2500	0.1824
1034631	Beta(2,2) baseline	9	7	0.1777	0.2500	0.1404
1034632	L5	8	7	0.1802	0.2793	0.1525
1034633	L5	9	7	0.2085	0.2791	0.1901
1034634	L5	8	7	0.1837	0.2821	0.1613
1034661	Beta(2,2) baseline	9	7	0.1835	0.2500	0.1429
1034668	L1c	9	7	0.1985	0.2500	0.1683
1034689	L1c	9	7	0.1789	0.2500	0.1352
1034696	Beta(2,2) baseline	9	7	0.2017	0.2500	0.1735
1034704	Beta(2,2) baseline	8	7	0.1790	0.2500	0.1426

F Training sources, pretrained models and compute

Training sources. All training and development data come from measurement-db at revision `54074453da7b9911be0f48e4c560be547bc84d21`, restricted to the 166 binary benchmarks of Section 3. No other data source was used for training, tuning or simulation. The test weights use aggregate statistics of our own formative feedback (Section 5). Earlier entries V2 and V3 used four and five benchmarks of the same inventory [2].

Pretrained models. No model was fine-tuned, and no paid API was used. Table 6 lists every pretrained model we ran.

Table 6: **Pretrained models used in this work.** Only E5 runs inside the submission.

Model	Revision	Use
`intfloat/multilingual-e5-small`	`614241f622f53c4eeff9890bdc4f31cfecc418b3`	frozen encoder in training and in the submission
`Qwen/Qwen3-Embedding-0.6B`	`97b0c614be4d77ee51c0cef4e5f07c00f9eb65b3`	encoder comparison only
`Qwen/Qwen3-4B`	`1cfa9a7208912126459214e8b04321603b3df60c`	feature extraction, stopped before it finished
`Qwen/Qwen3-8B`	`b968826d9c46dd6066d109eabc6255188de91218`	difficulty judge

Compute. GPU work ran on Google Colab L4 instances, and one Colab CPU runtime timed the encoder. Together they used 25.9 compute units of our own. Corpus building, simulation, replays, packaging and this report ran on an Apple M4 laptop.